Predefining an analysis plan

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Predefining an analysis plan

Preserve the original plan, denominator and visible reasons for later changes.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of predefining an **analysis plan**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of privacy and data governance

Reference example, not the only acceptable wording: Teacher explanation: Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of dataset and benchmark limitations

Reference example, not the only acceptable wording: Teacher explanation: A benchmark measures performance on its particular data and labels. Duplicates, obsolete cases or poor coverage can weaken claims.

Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of multiple comparisons

Reference example, not the only acceptable wording: Teacher explanation: Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

For this exercise use Bonferroni:family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.

Pictured end goal

Explain predefining an **analysis plan** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Privacy and data governance

Dataset and benchmark limitations

Multiple comparisons

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. An **analysis plan** records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected. **Deviation**s remain visible.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of predefining an **analysis plan**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

An **analysis plan** records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected. **Deviation**s remain visible.

Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.

Plan item	Complete supplied exercise contract
Main population of cases	All twenty included by the original plan
Later proposed change	Drop four failed cases because they make the graph look worse
Known outcomes	Four proposed exclusions are failures; outcomes of the other sixteen are not supplied
Original denominator	Twenty; use actual pass count, not an invented sixteen passes
Picture rule	Twelve cases; exclude only unreadable files identified before scoring; report all versions
Picture change	Two readable model failures proposed for exclusion
Scope	Fictional audit; no actual timestamp, preregistration or experiment is created

Main evidence ledger and checked key

Main record	Supported statement	Unsupported shortcut
Original plan	All twenty cases included	Reconstruct eligibility after seeing outcomes
Four cases fail	Failure is known for those four	Assume every remaining case passed
Remove because graph looks worse	Outcome-informed exclusion violates original plan	Call it a predeclared eligibility rule
Original report	Actual passes / 20; numerator not supplied	Report 16/20 as an observed result
Conditional illustration	If all other sixteen passed, 16/20 = 80%	Treat the if-clause as evidence
Necessary revision	Retain original record and disclose revision/reason/timing	Erase old plan or present revised analysis as original

Main: exclusion of four failed cases because the graph looks worse is outcome-informed. Original denominator remains twenty; the other sixteen outcomes are unknown, so 16/20 is justified only conditionally if all sixteen passed. Picture: readable model failures do not meet the predeclared unreadable-file rule; keep original results and disclose any later deviation.

Independent case: fifteen inputs, three predeclared malformed exclusions, twelve eligible and nine passes gives 9/12 = 75% eligible pass rate and coverage 12/15 = 80%. The 9/15 = 60% all-input rate is a different metric. Keep excluded IDs/rule. No actual preregistration or valid-assumption guarantee follows.

Predefining an analysis plan — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse.	Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.	The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse.

The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Plan audit	Checked interpretation
All twenty cases planned	Four unattractive failures stay in original analysis
Sixteen other cases remain	Their pass/fail outcomes are unknown
Change disclosed	Makes change visible; does not remove all resulting bias
Version label or timestamp	Metadata alone does not prove chronology
Predeclared exclusion truly met	Report excluded IDs, rule and coverage
Failures are readable	Not eligible under unreadable-file rule
Analysis plan exists	Sampling and test assumptions still require justification

Separate pictured facts and full reasoning

Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

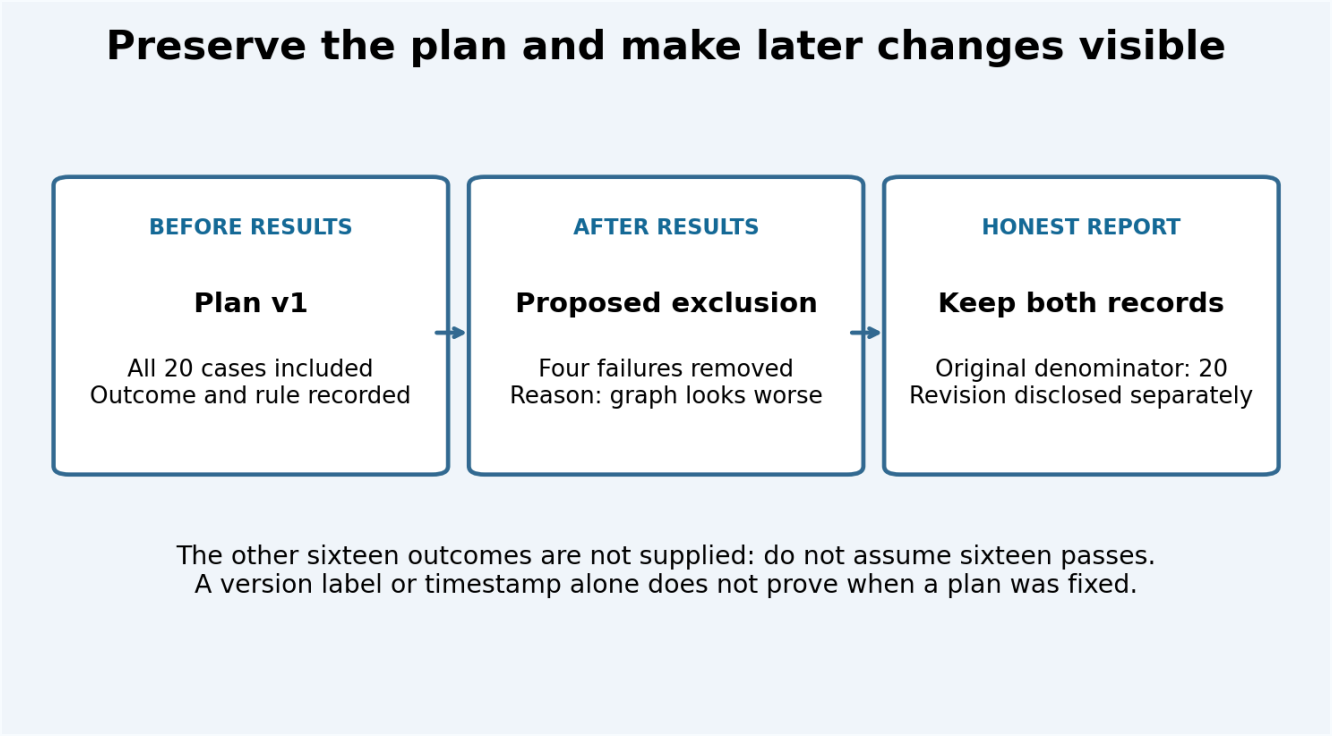
Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

The separate picture records a twelve-case plan before scoring. Only unreadable files identified before scoring are eligible for exclusion; all version results are reported. A readable case on which the model fails is not unreadable. Removing two such cases after seeing failures violates this rule.

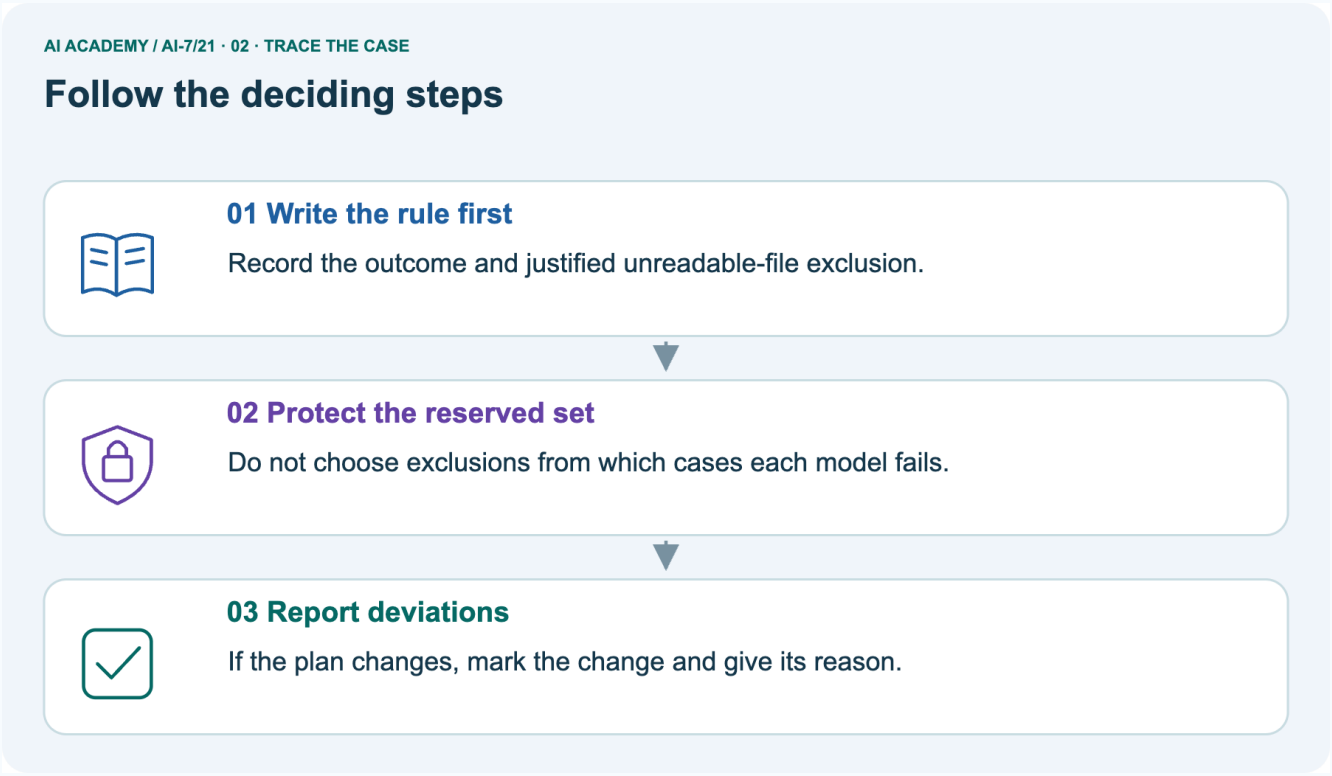
The picture supplies no actual unreadable-file count or pass count, so no twelve-case performance rate can be calculated. A later necessary correction must preserve the original plan and analysis and be identified as a deviation, with reasons and timing. This fictional

account does not prove a real plan was fixed before outcomes.

Picture evidence	Rule application
Twelve planned cases	Starting set, not a supplied twelve-case pass count
Unreadable before scoring	Exclusion allowed by this stated rule
Readable model failure	Does not meet unreadable-file exclusion
Two readable failures removed	Outcome-informed deviation; retain original analysis
Actual unreadable count / actual passes	Not supplied; no invented rate
Documented revision	Transparent change, not proof the original design was valid



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write the rule first	Record the outcome and justified unreadable-file exclusion.
2. Protect the reserved set	Do not choose exclusions from which cases each model fails.
3. Report deviations	If the plan changes, mark the change and give its reason.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **analysis plan** limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

Change: Exclude two readable cases because the new model fails them.

Prediction: That outcome-driven exclusion violates the recorded rule.

Reason: Model failure is not the predeclared unreadable-file criterion.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Analysis plan	Declared procedures before inspecting primary results
Exclusion rule	A stated criterion for removing an ineligible case
Deviation	A recorded departure from the original plan

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

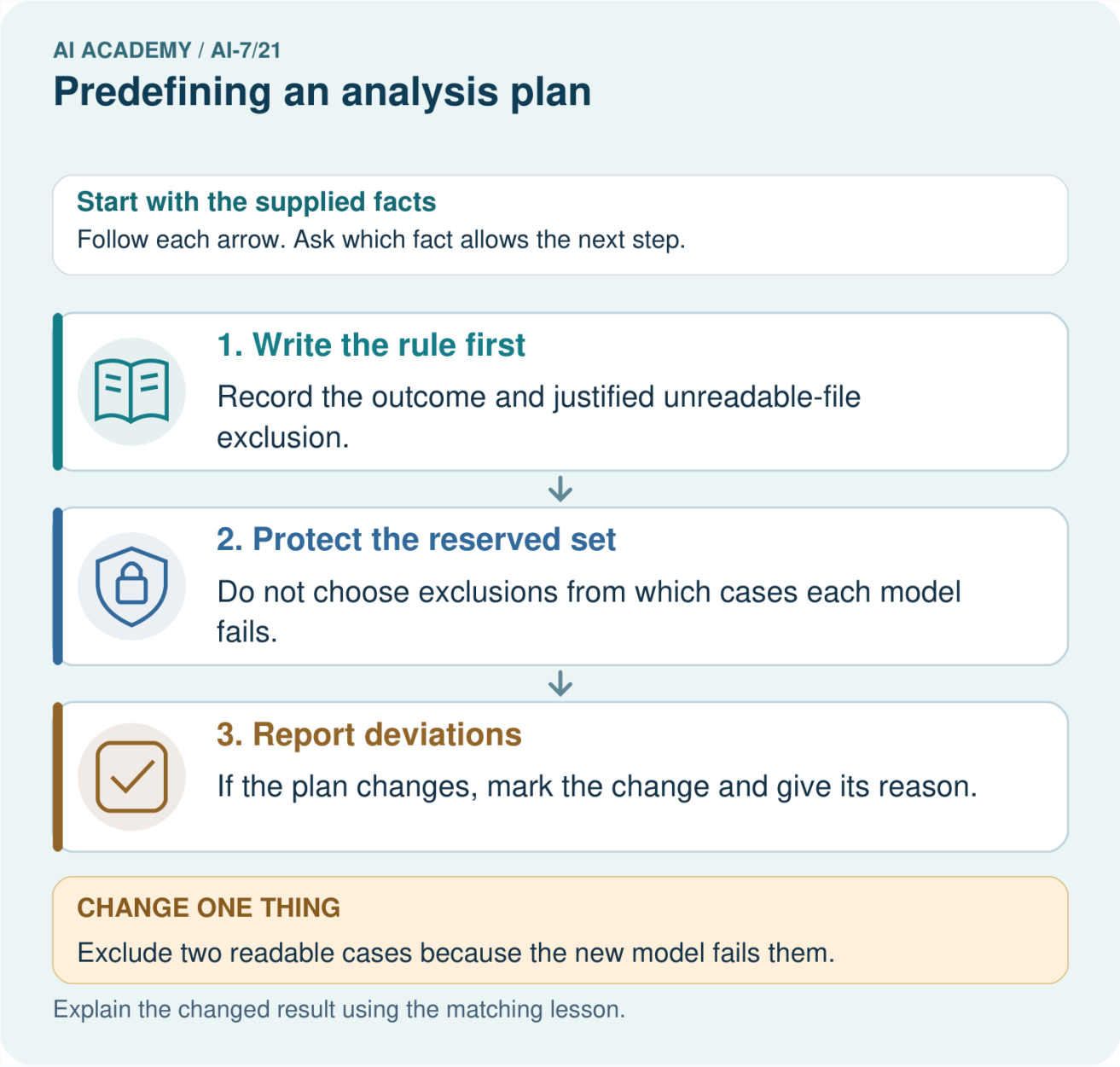
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Predefining an analysis plan

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Predefining an **analysis plan**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

Record the outcome and justified unreadable-file exclusion. Do not choose exclusions from which cases each model fails. If the plan changes, mark the change and give its reason.

Changed case: That outcome-driven exclusion violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

Changed-case reasoning: That outcome-driven exclusion violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main: exclusion of four failed cases because the graph looks worse is outcome-informed. Original denominator remains twenty; the other sixteen outcomes are unknown, so 16/20 is justified only conditionally if all sixteen passed. Picture: readable model failures do not meet the predeclared unreadable-file rule; keep original results and disclose any later deviation.

Independent case: fifteen inputs, three predeclared malformed exclusions, twelve eligible and nine passes gives 9/12 = 75% eligible pass rate and coverage 12/15 = 80%. The 9/15 = 60% all-input rate is a different metric. Keep excluded IDs/rule. No actual preregistration or valid-assumption guarantee follows.

Diagnose the current misconceptions

The worked result establishes every future case: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Plan excludes malformed **input**s by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

Teacher explanation: Eligible pass 9/12=75%; coverage 12/15 after three prespecified exclusions. Preserve excluded IDs and rule; do not call 9/15 the same metric.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: An **analysis plan** records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected. **Deviation**s remain visible.

Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.

W2 Trace the worked case

Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

W3 Audit the limit

A report omits this qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Plan excludes malformed **input**s by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

Teacher explanation: Eligible pass 9/12=75%; coverage 12/15 after three prespecified exclusions. Preserve excluded IDs and rule; do not call 9/15 the same metric.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Eligible pass 9/12=75%; coverage 12/15 after three prespecified exclusions. Preserve excluded IDs and rule; do not call 9/15 the same metric. Require intermediate reasoning and the stated scope. Fresh retry: Ten eligible, eight completed, six pass; plan reports completed rate and missing count. Expected reasoning: 6/8=75% completed; two missing explicit. Preserve the planned denominator policy. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled predefining an **analysis plan evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of predefining an **analysis plan**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Pilot experiments. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/22, ai-7/24, ai-7/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of predefining an **analysis plan**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Check every learner before discussion: Use the worked case for Predefining an **analysis plan**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Exclude two readable cases because the new model fails them. Require a **prediction** and a reason before comparing. Expected result: That outcome-driven exclusion violates the recorded rule.

Reason: Model failure is not the predeclared unreadable-file criterion.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

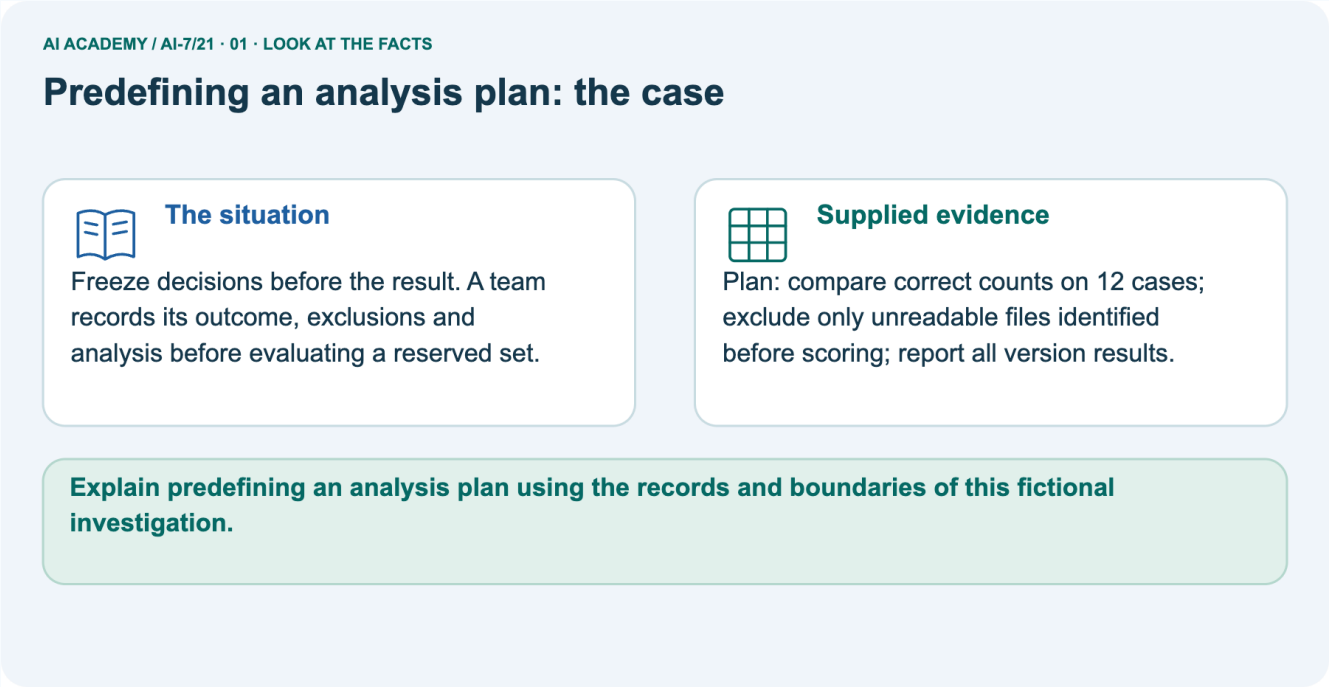
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Predefining an analysis plan: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

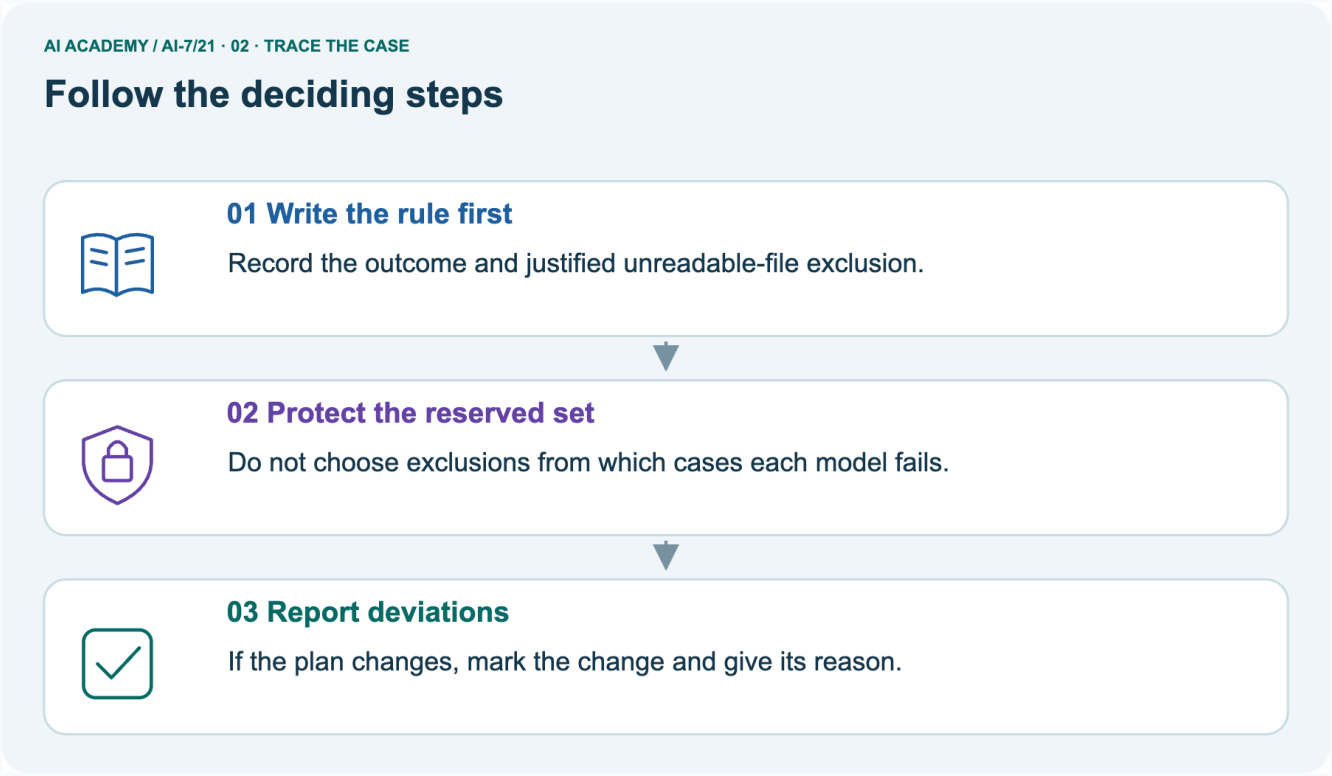
Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Supplied facts: Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



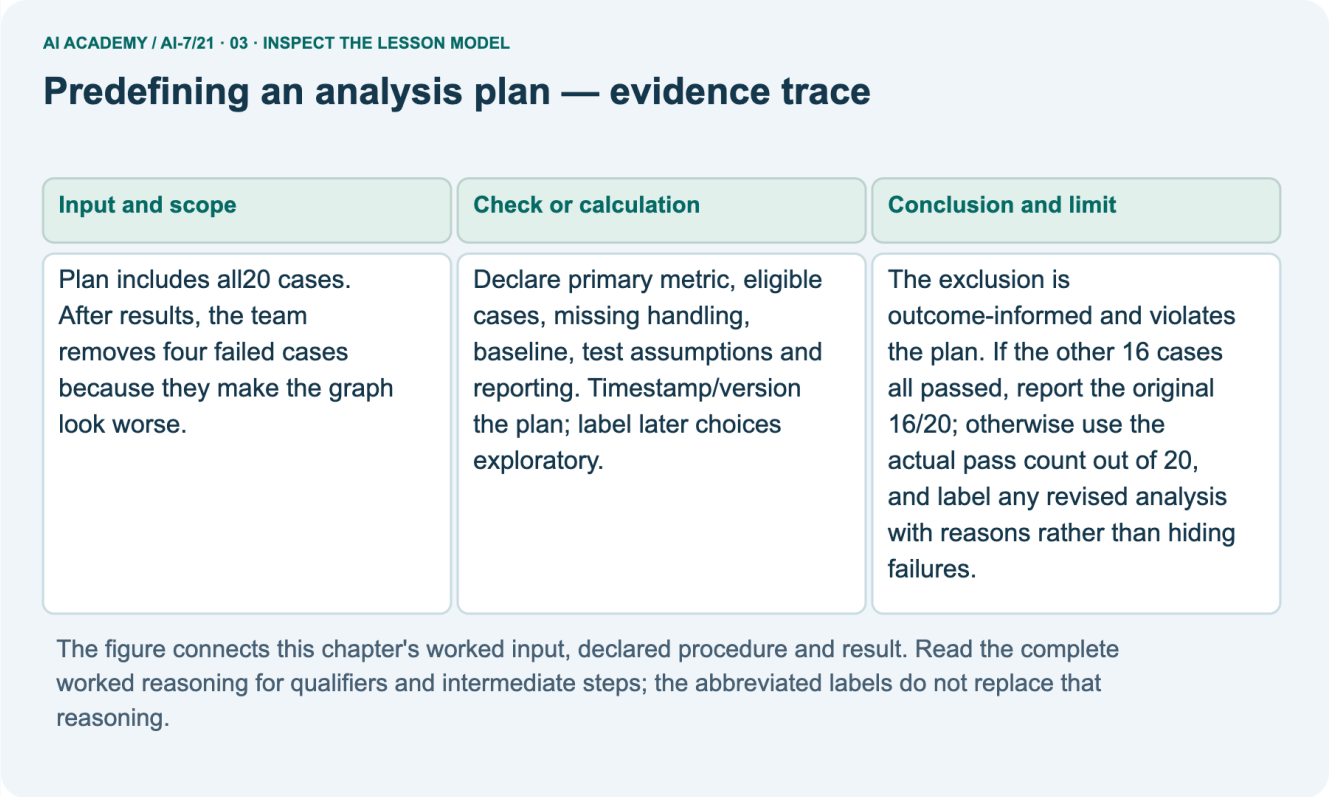
Follow the labelled picture with the matching explanation. Open the vector diagram

Write the rule first: Record the outcome and justified unreadable-file exclusion. → Protect the reserved set: Do not choose exclusions from which cases each model fails. → Report **deviation**s: If the plan changes, mark the change and give its reason.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Predefining an analysis plan — evidence trace



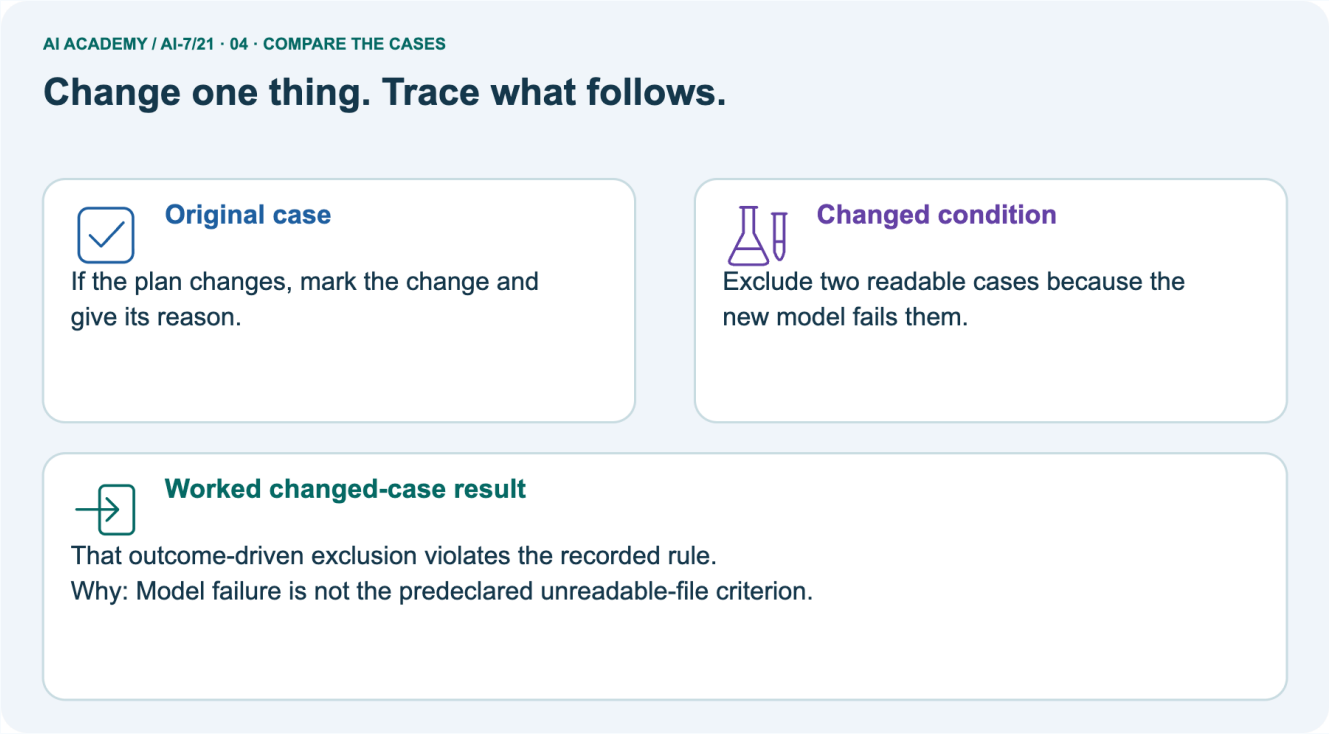
Follow the labelled picture with the matching explanation. Open the vector diagram

Predefining an **analysis plan** — **evidence** trace. Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. | Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory. | The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If the plan changes, mark the change and give its reason. Changed condition: Exclude two readable cases because the new model fails them.

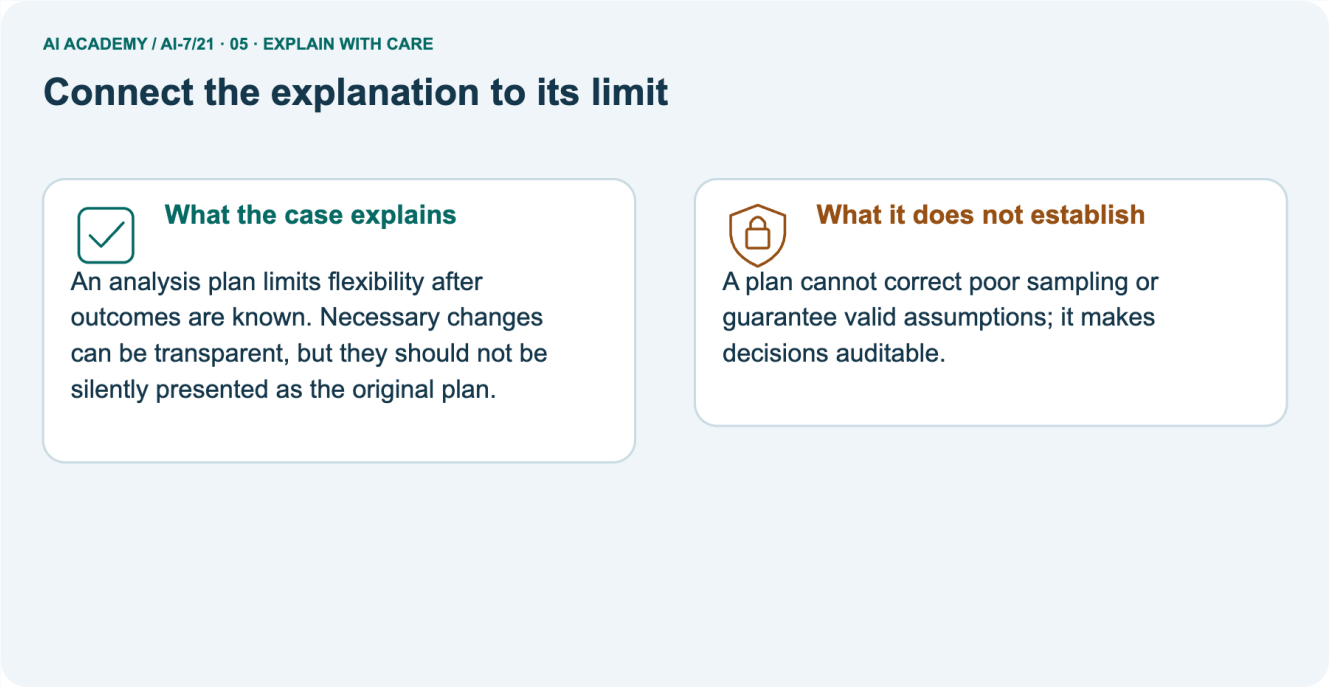
Expected result: That outcome-driven exclusion violates the recorded rule. Reason: Model failure is not the predeclared unreadable-file criterion.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Model failure is not the predeclared unreadable-file criterion.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **analysis plan** limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

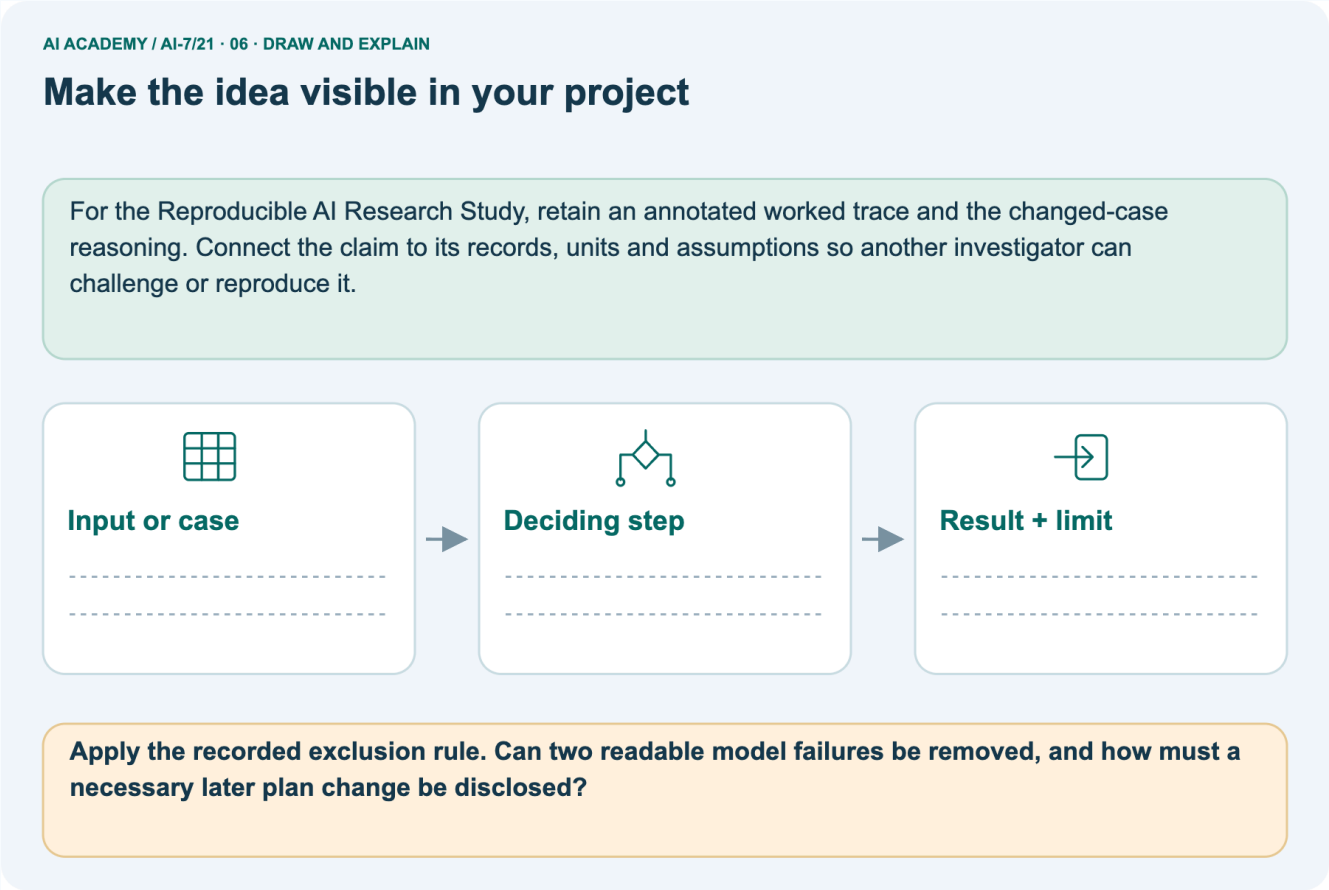
Limit: A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Record the outcome and justified unreadable-file exclusion. Do not choose exclusions from which cases each model fails. If the plan changes, mark the change and give its reason.

Changed case: That outcome-driven exclusion violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Predefining an analysis plan: the case

02. Follow the deciding steps

03. Predefining an analysis plan — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.